

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [115, 121, 118, 102, 100, 112, 97, 115, 108, 97]

Tap amplitudes (pixels): [139.08630414242805, 141.3223266154361, 148.48568954616468, 150.71164520368026, 142.6954799564443, 150.3761949245957, 139.5564401953561, 148.92279879185725, 138.97481786280562, 139.5313584826006, 139.08630414242805]

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Analysis

- Tap count: 11 taps detected; 10 periods provided, which is consistent (periods are between consecutive taps).
- - Periods (frames): [115, 121, 118, 102, 100, 112, 97, 115, 108, 97]
 - Variability is modest; no obvious long arrest/freeze (no single period stands out as dramatically larger than the rest).
 - Trend: Slight speeding up around periods 7 and 10 (97), otherwise roughly stable. This suggests no marked slowing; if anything, stable-to-slightly faster, so no penalty for slowing.
 - Interruptions/hesitations would appear as markedly prolonged periods or irregular spikes; none are apparent. This argues against multiple interruptions (>2).
- - Amplitudes (pixels): [139.09, 141.32, 148.49, 150.71, 142.70, 150.38, 139.56, 148.92, 138.97, 139.53, 139.09]
 - First 2–3 taps increase to a peak at tap 4–6, then several taps show a return to lower values (~139–143) interleaved with larger ones (taps 8: 148.9).
 - Decrement pattern: Not a monotonic decline from tap 2 onward (which would support “starting after the 1st tap”). Instead, there is an early buildup to peak, then variable amplitudes with multiple returns to baseline-sized amplitudes (~139) from tap 7 onward. This looks like slight-to-mild decrement near the latter half rather than a consistent early decrement.

- Because there are rebounds to high amplitude (taps 6 and 8), the decrement isn't sustained midway; it's more like slight variability with some smaller taps later.

Interpretation mapped to MDS-UPDRS 3.4 criteria:

- Interruptions/hesitations: None clearly evident (no 3–5 or >5 interruptions; no longer arrest).
- Speed: Not clearly slowed; periods are relatively uniform with some shorter values, arguing against mild or moderate slowing.
- Amplitude decrement: Some reduction in several later taps but not consistent starting after the 1st tap, and not a clear sustained mid-sequence decrement. This aligns better with “amplitude decrements near the end” (criterion 1c) rather than midway (2c) or after the 1st tap (3c).

Overall, this best fits “Slight (1)” due to minor late-sequence amplitude reduction without clear slowing or multiple interruptions.

Score: 1
